

TOPICS IN MATHEMATICAL SCIENCE

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INTRODUCTION TO KOSZUL DUALITY

AARON CHAN

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Convention

Throughout the course, $\mathbb{k}$ is a field. All rings are unital and associative. All modules are right modules unless otherwise specified. When we discuss dimensions, we mean dimensions as $\mathbb{k}$ -vector spaces.

1 Lecture 1: Graded rings and modules

Algebras and semisimplicity

Definition 1.1. A $\mathbb{k}$ -algebra is a $\mathbb{k}$ -vector space with an associative unital multiplication that is *bilinear*, meaning linear in each variable.

A right A -module is understood to be a $\mathbb{k}$ -vector space M with a bilinear action satisfying $(ma)b = m(ab)$ and $m1_A = m$ for $m \in M$ and $a, b \in A$.

Example 1.2. The matrix algebra $\text{Mat}_n(\mathbb{k})$ and its upper triangular subalgebra $\mathbb{k}A_n$ are $\mathbb{k}$ -algebras. For the matrix units e_{ij} , multiplication is $e_{ij}e_{kl} = \delta_{jk}e_{il}$. Each row space $e_{ii}\mathbb{k}A_n$ is a right $\mathbb{k}A_n$ -module.

Definition 1.3. A module L is *simple* if $L \neq 0$ and its only submodules are 0 and L . A module is *semisimple* if it is a direct sum of simple modules. An algebra B is *semisimple* if its right regular module B_B is semisimple. The *Jacobson radical* $\text{rad } B$ is the intersection of all maximal right ideals of B .

We use the following standard facts without proof. A *division $\mathbb{k}$ -algebra* is a non-zero $\mathbb{k}$ -algebra whose non-zero elements are invertible.

Theorem 1.4 (Artin–Wedderburn). For a non-zero finite-dimensional $\mathbb{k}$ -algebra B ,

$$B \text{ is semisimple} \iff \text{rad } B = 0 \iff B \cong \prod_{i=1}^r \text{Mat}_{n_i}(D_i),$$

where $r, n_i \geq 1$ and each D_i is a finite-dimensional division $\mathbb{k}$ -algebra. If $\mathbb{k}$ is algebraically closed, then $D_i = \mathbb{k}$ for every i . For semisimple B , every finite-dimensional left or right B -module is semisimple, and every submodule has a complement.

When all $D_i = \mathbb{k}$, the algebra is called *split semisimple*: it is a product of independent matrix blocks. For example, $\text{Mat}_2(\mathbb{k}) \times \text{Mat}_3(\mathbb{k})$ is the algebra of block-diagonal matrices with blocks of sizes 2 and 3. We do not require semisimple algebras to be split.

Example 1.5. For $S = \mathbb{k}^{\oplus r}$ with coordinate idempotents $e_1, \dots, e_r$,

$$1 = \sum_{i=1}^r e_i, \quad e_i e_j = \delta_{ij} e_i, \quad M = \bigoplus_{i=1}^r M e_i \quad \text{for every right } S\text{-module } M.$$

Thus S is semisimple. In contrast, $\text{rad}(\mathbb{k}A_2) = \mathbb{k}e_{12} \neq 0$.

Tensor products of vector spaces

Definition 1.6. For vector spaces U, V , the **tensor product** $U \otimes_{\mathbb{k}} V$ is equipped with a bilinear map $(u, v) \mapsto u \otimes v$ such that every bilinear map $b : U \times V \rightarrow W$ has a unique linear factorisation:

$$\begin{array}{ccc} U \times V & \xrightarrow{(u,v) \mapsto u \otimes v} & U \otimes_{\mathbb{k}} V \\ & \searrow b & \downarrow \exists! \tilde{b} \\ & & W. \end{array}$$

If $(u_i)_{i=1}^r$ and $(v_j)_{j=1}^s$ are bases, then $(u_i \otimes v_j)_{i,j}$ is a basis of $U \otimes_{\mathbb{k}} V$; hence $\dim_{\mathbb{k}}(U \otimes_{\mathbb{k}} V) = rs$. We use the canonical identifications $(U \otimes_{\mathbb{k}} V) \otimes_{\mathbb{k}} W \cong U \otimes_{\mathbb{k}} (V \otimes_{\mathbb{k}} W)$ and $\mathbb{k} \otimes_{\mathbb{k}} U \cong U$, and write $U^{\otimes 0} = \mathbb{k}$.

Graded algebras

Definition 1.7. Let G be a group with identity e . A **G -graded $\mathbb{k}$ -algebra** is a $\mathbb{k}$ -algebra A with $1_A \in A_e$ and a decomposition

$$A = \bigoplus_{g \in G} A_g \quad \text{such that} \quad A_g A_h \subset A_{gh} \quad \text{for all } g, h \in G.$$

An element of A_g is **homogeneous of degree g** .

Every $a \in A$ is a finite sum of elements of the spaces A_g , called its **homogeneous components**.

Example 1.8. Choose $g_1, \dots, g_n \in G$ and grade $\text{Mat}_n(\mathbb{k})$ by

$$A_h = \text{span}_{\mathbb{k}}\{e_{ij} \mid g_i^{-1} g_j = h\} \quad (h \in G).$$

The rule $e_{ij} e_{j\ell} = e_{i\ell}$ respects degrees because $(g_i^{-1} g_j)(g_j^{-1} g_\ell) = g_i^{-1} g_\ell$. For $G = \mathbb{Z}$ and $g_i = i$, we obtain $\deg e_{ij} = j - i$; the full matrix algebra then has negative as well as positive degrees.

Convention. Unless stated otherwise, a **graded algebra** means a $\mathbb{Z}$ -graded algebra concentrated in non-negative degrees:

$$A = \bigoplus_{i \in \mathbb{Z}} A_i, \quad A_i = 0 \text{ for } i < 0, \quad A_i A_j \subset A_{i+j}.$$

Example 1.9. For the upper triangular algebra $\mathbb{k}A_n$ of Example 1.2, set

$$(\mathbb{k}A_n)_d = \text{span}_{\mathbb{k}}\{e_{ij} \mid j - i = d\}, \quad S = (\mathbb{k}A_n)_0 = \bigoplus_{i=1}^n \mathbb{k}e_{ii} \cong \mathbb{k}^{\oplus n}.$$

Since $e_{ij} e_{k\ell} = \delta_{jk} e_{i\ell}$, multiplication adds degrees. For $n = 2$, the left and right actions on e_{12} differ:

$$e_{11} e_{12} = e_{12} = e_{12} e_{22}, \quad e_{22} e_{12} = e_{12} e_{11} = e_{12}^2 = 0.$$

Remark 1.10. The algebra $\mathbb{k}A_n$ is the path algebra of a linearly oriented type A_n quiver. The matrix-unit calculations in Example 1.9 express concatenation of composable paths; no quiver terminology is needed for the calculations.

Example 1.11. The algebra $\mathbb{k}A_3$ also has a $\mathbb{Z}^{\oplus 2}$ -grading with

$$\deg e_{ii} = (0, 0), \quad \deg e_{12} = (1, 0), \quad \deg e_{23} = (0, 1), \quad \deg e_{13} = (1, 1).$$

The homomorphism $\mathbb{Z}^{\oplus 2} \rightarrow \mathbb{Z}$, $(p, q) \mapsto p + q$, gives the grading of Example 1.9. Alternatively, setting $\deg e_{ij} = 2(j - i)$ gives a $\mathbb{Z}$ -grading with $A_1 = 0$ but $A_2 \neq 0$.

Example 1.12. Every algebra B has the *trivial grading*: $B_0 = B$ and $B_i = 0$ for $i \neq 0$. Thus a grading alone imposes no restriction on B_0 .

Definition 1.13. A *homomorphism of graded algebras* $f : A \rightarrow B$ is a $\mathbb{k}$ -algebra homomorphism with $f(A_i) \subset B_i$ for all $i \in \mathbb{Z}$.

Definition 1.14. A graded algebra A is *locally finite* if $\dim_{\mathbb{k}} A_i < \infty$ for every $i \in \mathbb{Z}$.

Assumption 1.15. For quadratic and Koszul algebras, $S := A_0$ is finite-dimensional semisimple.

For $\mathbb{k}A_n$, the degree zero algebra is $\mathbb{k}^{\oplus n}$; for $\text{Mat}_n(\mathbb{k})$ with the trivial grading, it is $\text{Mat}_n(\mathbb{k})$. Both satisfy Assumption 1.15. Definitions of graded algebras and modules do not require Assumption 1.15.

Graded modules

Definition 1.16. Let $A = \bigoplus_{g \in G} A_g$ be a G -graded algebra. A *G -graded right A -module* is a right A -module M with a direct sum decomposition

$$M = \bigoplus_{g \in G} M_g \quad \text{such that} \quad M_g A_h \subset M_{gh} \quad \text{for all } g, h \in G.$$

An element of M_g is called *homogeneous of degree g* .

For the $\mathbb{Z}$ -grading used in the course, this means

$$M = \bigoplus_{j \in \mathbb{Z}} M_j, \quad M_j A_i \subset M_{i+j} \quad \text{for all } i, j \in \mathbb{Z}.$$

We allow negative degrees for modules because shifts are unavoidable, even when the algebra is concentrated in non-negative degrees.

Example 1.17. The decomposition $A = \bigoplus_{i \in \mathbb{Z}} A_i$ makes the right regular module A_A a graded module. More generally, $A^{\oplus r}$ is a graded free module with each basis vector placed in degree 0.

Definition 1.18. Let M be a graded A -module and let $d \in \mathbb{Z}$. The *shift* $M(d)$ is the graded A -module whose underlying A -module is M , but whose grading is

$$M(d)_j = M_{j+d}.$$

By Definition 1.18, $1 \in A_0 = A(-d)_d$, so $A(-d)$ has a generator in degree d .

Example 1.19. Let $A = \mathbb{k}A_3$ and $P = e_{11}A$ with the grading of Example 1.9. The shifted module $P(-2)$ has basis

$$e_{11}, e_{12}, e_{13} \quad \text{in degrees } 2, 3, 4, \quad \text{respectively.}$$

Definition 1.20. Let M, N be graded A -modules. A homomorphism $f : M \rightarrow N$ is *homogeneous of degree d* if

$$f(M_j) \subset N_{j+d} \quad \text{for all } j \in \mathbb{Z}.$$

A *homomorphism of graded modules* means a homogeneous A -module homomorphism of degree 0.

Example 1.21. For $A = \mathbb{k}A_3$, left multiplication by e_{12} gives a right A -module homomorphism $e_{22}A \rightarrow e_{11}A$ of degree 1. Equivalently,

$$e_{22}A(-1) \longrightarrow e_{11}A, \quad u \longmapsto e_{12}u \quad \text{has degree 0.}$$

Right A -linearity is the identity $e_{12}(ua) = (e_{12}u)a$ for $u \in e_{22}A$ and $a \in A$.

Homogeneous ideals and quotients

Definition 1.22. Let $A = \bigoplus_{i \in \mathbb{Z}} A_i$ be a graded algebra. A right ideal $I \subset A$ is **homogeneous** if

$$I = \bigoplus_{i \in \mathbb{Z}} (I \cap A_i).$$

Equivalently, whenever $a \in I$ and $a = a_0 + \cdots + a_N$ is its decomposition into homogeneous components, each a_i also belongs to I .

For quotient algebras we need two-sided ideals. If I is a homogeneous two-sided ideal, then

$$A/I = \bigoplus_{i \in \mathbb{Z}} A_i / (I \cap A_i)$$

is again a graded algebra.

Example 1.23. In $A = \mathbb{k}A_3$ with the grading of Example 1.9, the two-sided ideal $I = (e_{13}) = \mathbb{k}e_{13}$ is homogeneous. Writing $\bar{e}_{ij}$ for the image of e_{ij} ,

$$(A/I)_0 \cong \mathbb{k}^{\oplus 3}, \quad (A/I)_1 = \mathbb{k}\bar{e}_{12} \oplus \mathbb{k}\bar{e}_{23}, \quad (A/I)_d = 0 \quad (d \geq 2).$$

In particular, $\bar{e}_{12}\bar{e}_{23} = 0$.

Example 1.24. In $A = \mathbb{k}A_2$, the subspace $I = \mathbb{k}(e_{22} + e_{12})$ is a right ideal: $Ie_{11} = Ie_{12} = 0$ and $Ie_{22} = I$. However, $e_{22}, e_{12} \notin I$, so I is not homogeneous. The right module A/I therefore does not inherit the quotient grading.

Definition 1.25. Let A be a graded algebra and set

$$A_{>0} := \bigoplus_{i>0} A_i.$$

The subspace $A_{>0}$ is a homogeneous two-sided ideal, called the **positive degree ideal**.¹ The quotient $A/A_{>0} \cong A_0 =: S$ is called the **degree zero quotient**.

The quotient map

$$\pi : A \longrightarrow S, \quad \pi \left(\sum_{i \in \mathbb{Z}} a_i \right) = a_0,$$

makes S into both a left and a right A -module, with commuting actions.

The degree zero quotient is the basic module in the definition of a Koszul algebra. Every positive degree element acts by zero, but S may have dimension greater than one and may itself be non-commutative.

Hilbert series

Definition 1.26. Let $M = \bigoplus_{j \in \mathbb{Z}} M_j$ be a graded vector space with $\dim_{\mathbb{k}} M_j < \infty$ for all j and $M_j = 0$ for all $j \ll 0$. The **Hilbert series** of M is the formal Laurent series

$$h_M(t) := \sum_{j \in \mathbb{Z}} (\dim_{\mathbb{k}} M_j) t^j.$$

For a locally finite graded algebra A , we write $h_A(t)$ for its Hilbert series as a graded vector space.

¹The ideal $A_{>0}$ is also called the augmentation ideal, relative to the projection $A \rightarrow A_0$. We avoid this terminology in these notes.

The Hilbert series records the size of each degree. It forgets multiplication, but it is still strong enough to give numerical tests for Koszulness later.

Lemma 1.27. *Let M, N be locally finite graded vector spaces concentrated in degrees bounded below.*

(1) *If $M \cong N$ as graded vector spaces, then $h_M(t) = h_N(t)$.*

(2) *If $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$ is an exact sequence of graded vector spaces with degree 0 maps, then*

$$h_M(t) = h_L(t) + h_N(t).$$

(3) *For any shift $d \in \mathbb{Z}$,*

$$h_{M(d)}(t) = t^{-d} h_M(t).$$

Proof All three statements follow by taking the degree j part. For example, $M(d)_j = M_{j+d}$, so

$$h_{M(d)}(t) = \sum_j \dim_{\mathbb{k}}(M_{j+d}) t^j = t^{-d} \sum_i \dim_{\mathbb{k}}(M_i) t^i.$$

□

Example 1.28. *For the upper triangular matrix algebra $\mathbb{k}A_n$,*

$$h_{\mathbb{k}A_n}(t) = n + (n-1)t + (n-2)t^2 + \cdots + t^{n-1}.$$

The coefficient of t^d counts the matrix units e_{ij} with $j - i = d$.

Example 1.29. *For $A = \mathbb{k}A_3$ and $I = (e_{13})$ from Example 1.23,*

$$h_A(t) = 3 + 2t + t^2, \quad h_I(t) = t^2, \quad h_{A/I}(t) = 3 + 2t.$$

Thus $h_A = h_I + h_{A/I}$.

Lemma 1.30. *If V, W are locally finite graded vector spaces concentrated in degrees bounded below, then*

$$h_{V \otimes W}(t) = h_V(t) h_W(t),$$

where $(V \otimes W)_n = \bigoplus_{i+j=n} V_i \otimes W_j$.

Proof The degree n part has dimension

$$\dim_{\mathbb{k}}(V \otimes W)_n = \sum_{i+j=n} \dim_{\mathbb{k}} V_i \dim_{\mathbb{k}} W_j.$$

This is exactly the coefficient of t^n in the product $h_V(t) h_W(t)$. □

Graded $\mathbb{k}$ -linear duals

A linear functional on a graded vector space need not have a degree: it may be non-zero on infinitely many homogeneous components. To obtain a graded vector space, we retain only finite sums of homogeneous functionals.

Definition 1.31. *For a vector space V , its $\mathbb{k}$ -linear dual is $D(V) := \text{Hom}_{\mathbb{k}}(V, \mathbb{k})$. For a graded vector space $V = \bigoplus_{i \in \mathbb{Z}} V_i$, its **graded $\mathbb{k}$ -linear dual** is*

$$\mathbb{D}(V) := \bigoplus_{i \in \mathbb{Z}} D(V_{-i}), \quad \text{with} \quad \mathbb{D}(V)_i = D(V_{-i}).$$

A functional on V_{-i} extends by zero on the other components and has degree i . Thus $\mathbb{D}(V) \subset D(V)$, whereas $D(V) \cong \prod_{i \in \mathbb{Z}} D(V_i)$. These spaces coincide when V is finite-dimensional. Local finiteness alone does not make them equal.

Linear duality reverses the side of a module action. For a right A -module M and a left A -module N , define

$$(af)(m) := f(ma), \quad (ga)(n) := g(an) \quad (a \in A, f \in D(M), g \in D(N)). \quad (1.1)$$

Here $m \in M$ and $n \in N$; the first formula makes $D(M)$ a left module and the second makes $D(N)$ a right module. For graded modules, the same formulas preserve the graded $\mathbb{k}$ -linear dual: if $a \in A_p$ and $f \in \mathbb{D}(M)_q$, then af vanishes outside M_{-p-q} and hence has degree $p+q$. There are no signs in these actions.

Example 1.32. Let $A = \mathbb{k}A_3$ and $P = e_{11}A$ as in Example 1.19. The functionals $\varepsilon_{1j}(e_{1\ell}) = \delta_{j\ell}$ for $1 \leq j, \ell \leq 3$ form a basis of $\mathbb{D}(P)$, with

$$\deg \varepsilon_{1j} = 1 - j, \quad e_{12}\varepsilon_{12} = \varepsilon_{11}, \quad e_{23}\varepsilon_{13} = \varepsilon_{12}, \quad e_{jj}\varepsilon_{1j} = \varepsilon_{1j}.$$

The actions follow from (1.1). For comparison with an infinite-dimensional space, take $V = \bigoplus_{i \geq 0} \mathbb{k}v_i$ with $\deg v_i = i$. The functional $\lambda(v_i) = 1$ for every i belongs to $D(V)$ but not to $\mathbb{D}(V)$.

Lemma 1.33. On locally finite graded modules, $\mathbb{D}$ reverses arrows and short exact sequences, and evaluation gives a graded module isomorphism $M \cong \mathbb{D}(\mathbb{D}(M))$.

Proof A homomorphism $u : M \rightarrow N$ of degree 0 induces $\mathbb{D}(u) : \mathbb{D}(N) \rightarrow \mathbb{D}(M)$ by $f \mapsto f \circ u$. Dualising a short exact sequence in each degree is exact, since every linear functional on a subspace extends to the whole vector space. The evaluation map sends m to $(f \mapsto f(m))$. It is an isomorphism in each degree by finite-dimensional linear algebra, and (1.1) makes it a module homomorphism. $\square$

Left, right, and two-sided duals over S

Fix a finite-dimensional semisimple $\mathbb{k}$ -algebra S , concentrated in degree 0. Besides the $\mathbb{k}$ -linear dual, we need duals defined by left S -linear maps, right S -linear maps, and maps respecting both actions. Their definitions differ, even though the resulting spaces can be identified after a choice of form.

Write S^{op} for the [opposite algebra](#), with multiplication $s^{\text{op}}t^{\text{op}} = (ts)^{\text{op}}$. A right S -module is a left S^{op} -module. In the duality notation, Hom_S means left S -linear maps, and $\text{Hom}_{S^{\text{op}}}$ means right S -linear maps.

Definition 1.34. For a left S -module V and a right S -module W , define their [S-duals](#) by

$$\begin{aligned} V^* &:= \text{Hom}_S(V, S), & (fs)(v) &:= f(v)s, \\ {}^*W &:= \text{Hom}_{S^{\text{op}}}(W, S), & (sg)(w) &:= sg(w). \end{aligned}$$

Here $s \in S$, $v \in V$, $w \in W$, $f \in V^*$, and $g \in {}^*W$.

Thus V^* is a right S -module and *W is a left S -module. We use the notation of [BGS96, §2.7]; the position of the star records the resulting side.

Definition 1.35. An [S-bimodule](#) is a vector space M with $\mathbb{k}$ -bilinear left and right S -module actions satisfying

$$(sm)t = s(mt) \quad \text{for all } s, t \in S, m \in M.$$

The [enveloping algebra](#) of S is $S^{\text{en}} := S \otimes_{\mathbb{k}} S^{\text{op}}$, with multiplication

$$(s \otimes t^{\text{op}})(s' \otimes (t')^{\text{op}}) = ss' \otimes (t't)^{\text{op}} \quad (s, t, s', t' \in S).$$

An S -bimodule is equivalently a left S^{en} -module, with $(s \otimes t^{\text{op}})m = smt$. The two actions of $\mathbb{k}$ agree because the module actions are $\mathbb{k}$ -bilinear. All three spaces $D(M)$, M^* , and *M inherit bimodule structures:

$$(s\lambda t)(m) = \lambda(tms), \quad (sft)(m) = f(ms)t, \quad (sgt)(m) = sg(tm), \quad (1.2)$$

where $\lambda \in D(M)$, $f \in M^*$, $g \in {}^*M$, $s, t \in S$, and $m \in M$.

Definition 1.36. The *two-sided dual* of an S -bimodule M is

$$M^{*\text{en}} := \text{Hom}_{S^{\text{en}}}(M, S^{\text{en}}), \quad (sFt)(m) := F(m)(t \otimes s^{\text{op}}) \quad (s, t \in S, F \in M^{*\text{en}}, m \in M).$$

The output is initially a right S^{en} -module by multiplication on the values. The formula in Definition 1.36 converts that right action into an S -bimodule structure by interchanging the factors. In particular, the two-sided dual is not $\text{Hom}_{S^{\text{en}}}(M, S)$.

For comparison, the four notations in [GI20, §2.1] translate as follows:

these notes	$D(M)$ M^* *M $M^{*\text{en}}$
Grant–Iyama	M^* $M^{*\ell}$ M^{*r} $M^\vee$

The subscripts ℓ and r in their notation specify left and right linearity, respectively.

Comparing the duals

To compare an S -valued map with a $\mathbb{k}$ -valued map, we need a linear functional on S that retains all the information in its multiplication.

Definition 1.37. A *symmetrising form* on a finite-dimensional $\mathbb{k}$ -algebra S is a linear map $\tau : S \rightarrow \mathbb{k}$ satisfying

$$\tau(st) = \tau(ts) \quad \text{for all } s, t \in S, \quad (\tau(st) = 0 \text{ for all } t \in S) \implies s = 0.$$

An algebra admitting such a form is *symmetric*.

We use the standard theorem that every finite-dimensional semisimple $\mathbb{k}$ -algebra is symmetric; see [GI20, §2.1, p. 2593]. We do not prove this structure theorem here. The forms needed in our basic examples are explicit.

Example 1.38. For $S = \mathbb{k}^{\oplus r}$, take $\tau(a_1, \dots, a_r) = \sum_i a_i$. For $S = \text{Mat}_n(\mathbb{k})$, take $\tau(a) = \text{Tr}(a)$. Indeed, $\text{Tr}(ab) = \text{Tr}(ba)$ and $\text{Tr}(ae_{ji}) = a_{ij}$, so the pairing is non-degenerate in every characteristic. More generally, the sum of the matrix traces works on $\prod_{j=1}^r \text{Mat}_{n_j}(\mathbb{k})$, so it gives a symmetrising form for every split semisimple algebra in Theorem 1.4.

Proposition 1.39. Choose a symmetrising form τ on S . For every finite-dimensional S -bimodule M , there are isomorphisms of S -bimodules

$$\alpha_M : M^* \longrightarrow D(M), \quad f \longmapsto \tau \circ f, \quad \beta_M : {}^*M \longrightarrow D(M), \quad g \longmapsto \tau \circ g. \quad (1.3)$$

There is also an isomorphism $\gamma_M : M^{*\text{en}} \rightarrow {}^*M$ given by

$$\gamma_M(F)(m) = \sum_j \tau(a_j) b_j \quad \text{where} \quad F(m) = \sum_j a_j \otimes b_j^{\text{op}}. \quad (1.4)$$

The maps $\alpha_M, \beta_M, \gamma_M$ commute with precomposition by bimodule homomorphisms. The maps α_M and β_M also apply to left and right S -modules, respectively, with the opposite side on the duals.

Proof For $\lambda \in D(M)$, non-degeneracy of τ determines $f(m), g(m) \in S$ uniquely by

$$\tau(sf(m)) = \lambda(sm), \quad \tau(g(m)s) = \lambda(ms) \quad (s \in S, m \in M). \quad (1.5)$$

Equation (1.5) gives $f(sm) = sf(m)$, $g(ms) = g(m)s$, and $\alpha_M(f) = \beta_M(g) = \lambda$. Hence α_M, β_M are bijective. For bimodule compatibility, use (1.2) and $\tau(f(ms)t) = \tau(f(tms))$; the calculation for β_M uses $\tau(sg(tm)) = \tau(g(tms))$.

The form $\tau_{\text{en}}(a \otimes b^{\text{op}}) = \tau(a)\tau(b)$ is symmetrising on S^{en} . The formula (1.4) satisfies

$$\gamma_M(F)(ms) = \gamma_M(F)(m)s, \quad \beta_M(\gamma_M(F)) = \tau_{\text{en}} \circ F.$$

The argument (1.5), applied to S^{en} and τ_{en} , makes $F \mapsto \tau_{\text{en}} \circ F$ an isomorphism. Thus γ_M is an isomorphism. Composition with τ and τ_{en} commutes with precomposition, giving naturality. $\square$

The identifications (1.3) and (1.4) depend on τ . For instance, $\tau(a_1, \dots, a_r) = \sum_i c_i a_i$ is symmetrising on $\mathbb{k}^{\oplus r}$ for any $c_i \neq 0$. No semisimplicity assumption on S^{en} was used.

Definition 1.40. For a graded S -bimodule M , define its *graded S -duals* and *graded two-sided dual* by

$$M^{(*)} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^*, \quad ({}^*)M := \bigoplus_{i \in \mathbb{Z}} {}^*(M_{-i}), \quad M^{(*\text{en})} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^{*\text{en}},$$

where each indicated summand has degree i . The first construction also applies to graded left S -modules, and the second to graded right S -modules.

Parentheses on the stars distinguish the graded constructions from the full S -duals, as in [GI20, §2.1].

Corollary 1.41. For a locally finite graded S -bimodule M , a choice of symmetrising form gives natural isomorphisms of graded S -bimodules

$$M^{(*)} \cong \mathbb{D}(M) \cong ({}^*)M \cong M^{(*\text{en})}.$$

In particular, $\mathbb{D}(M) \cong M^{(*)}$ for a graded left S -module, and $\mathbb{D}(M) \cong ({}^*)M$ for a graded right S -module.

Proof Apply Proposition 1.39 to M_{-i} and take the direct sum over $i \in \mathbb{Z}$. Every comparison preserves degree because τ has degree 0. $\square$

For a graded A -module, $\mathbb{D}(M)$ has the opposite A -action in (1.1). An identification in Corollary 1.41 transports that action to the corresponding graded S -dual. These remain duals over S , not duals taking values in A .

Example 1.42. Let $S = \mathbb{k}^{\oplus r}$ with coordinate idempotents $e_1, \dots, e_r$ and the form $\tau(e_i) = 1$. For an S -bimodule M , the decompositions and the reversal of sides are

$$M = \bigoplus_{i,j} e_i M e_j, \quad e_j D(M) e_i \cong D(e_i M e_j) \cong e_j M^* e_i \cong e_j ({}^* M) e_i.$$

For the degree 1 piece $V = \mathbb{k}e_{12}$ of $\mathbb{k}A_2$, we have $e_1 V e_2 = V$. Its graded $\mathbb{k}$ -linear dual is spanned by ε of degree -1 , with $\varepsilon(e_{12}) = 1$ and $e_2 \varepsilon e_1 = \varepsilon$. The corresponding elements of V^* and *V send e_{12} to e_1 and e_2 , respectively. Thus equal scalar-valued functionals can correspond to different S -valued maps.

Lemma 1.43. For finite-dimensional left S -modules V and right S -modules W , evaluation gives canonical isomorphisms $V \cong (V^*)^*$ and $W \cong ({}^*W)^*$. For bimodules, both are bimodule isomorphisms. No symmetrising form is needed.

Proof By Theorem 1.4, every finite-dimensional S -module is a direct summand of some $S^{\oplus n}$. Evaluation is an isomorphism for S , hence for such summands; (1.2) gives bimodule compatibility. $\square$

2 Lecture 2: Tensor and quadratic algebras

Tensor products over an algebra

The degree zero algebra acts on both sides of the degree 1 piece. We therefore need the tensor product over an algebra, not only the tensor product over $\mathbb{k}$.

Definition 2.1. Let M be a right S -module, let N be a left S -module, and let W be a $\mathbb{k}$ -vector space. The **tensor product** $M \otimes_S N$ is the quotient of $M \otimes_{\mathbb{k}} N$ by the subspace generated by

$$ms \otimes n - m \otimes sn \quad (m \in M, n \in N, s \in S).$$

Equivalently, a bilinear map $b : M \times N \rightarrow W$ factors uniquely through a linear map $M \otimes_S N \rightarrow W$ precisely when $b(ms, n) = b(m, sn)$ for all $m \in M, s \in S$, and $n \in N$.

The equality $ms \otimes_S n = m \otimes_S sn$ allows a scalar from S to pass between adjacent factors. For an S -bimodule V , define

$$V^{\otimes_S 0} := S, \quad V^{\otimes_S i} := \underbrace{V \otimes_S \cdots \otimes_S V}_{i \text{ factors}} \quad (i \geq 1).$$

Tensor duality reverses the order of the factors. We record the four compatible forms used in [BGS96, §2.7] and [GI20, §2.1, (1)–(4)].

Proposition 2.2. Let S be finite-dimensional semisimple and let V, W be finite-dimensional S -bimodules. There are natural bimodule isomorphisms

$$\begin{aligned} \theta_D : W^* \otimes_S D(V) &\xrightarrow{\sim} D(V \otimes_S W), \\ \theta_\ell : W^* \otimes_S V^* &\xrightarrow{\sim} (V \otimes_S W)^*, \\ \theta_r : {}^*W \otimes_S {}^*V &\xrightarrow{\sim} {}^*(V \otimes_S W), \\ \theta_{\text{en}} : {}^*W \otimes_S V^{*\text{en}} &\xrightarrow{\sim} (V \otimes_S W)^{*\text{en}}, \end{aligned} \tag{2.1}$$

defined by

$$\begin{aligned} \theta_D(g \otimes \lambda)(v \otimes w) &= \lambda(vg(w)), \\ \theta_\ell(g \otimes f)(v \otimes w) &= f(vg(w)), \\ \theta_r(g \otimes f)(v \otimes w) &= g(f(v)w), \\ \theta_{\text{en}}(g \otimes F)(v \otimes w) &= \sum_j a_j \otimes g(b_j w)^{\text{op}}, \quad \text{where } F(v) = \sum_j a_j \otimes b_j^{\text{op}}. \end{aligned} \tag{2.2}$$

Proof The formulas in (2.2) respect the tensor relations and bimodule actions. The maps θ_D and θ_ℓ are isomorphisms for $W = S$ as a left module, hence for direct sums and direct summands of S . For θ_r , apply the same argument to the right module V . Choose a symmetrising form τ as in Definition 1.37, and use $\gamma_V, \gamma_{V \otimes_S W}$ from (1.4). Then

$$\gamma_{V \otimes_S W} \circ \theta_{\text{en}} = \theta_r \circ (\text{id}_{{}^*W} \otimes \gamma_V),$$

since both sides evaluate $g \otimes F$ at $v \otimes w$ as $\sum_j \tau(a_j)g(b_j w)$. Proposition 1.39 makes both γ maps isomorphisms, so θ_{en} is an isomorphism. $\square$

For a fixed symmetrising form τ , let $\alpha_M(f) = \tau \circ f$ as in (1.3). Then tensor evaluation satisfies

$$\alpha_{V \otimes_S W} \circ \theta_\ell = \theta_D \circ (\text{id}_{W^*} \otimes \alpha_V).$$

Both sides send $g \otimes f$ to $v \otimes w \mapsto \tau(f(vg(w)))$.

Example 2.3. Let $S = \mathbb{k}^{\oplus r}$ with coordinate idempotents $e_1, \dots, e_r$, and suppose $V = e_i V e_j$ and $W = e_j W e_\ell$. By (1.2),

$$V \otimes_S W = e_i(V \otimes_S W)e_\ell, \quad W^* \otimes_S V^* = e_\ell(W^* \otimes_S V^*)e_i.$$

Thus θ_ℓ in (2.1) reverses the two outer idempotents.

Generated algebras

Let A be graded with $S = A_0$. Multiplication makes A_1 an S -bimodule and gives

$$\mu_i : A_1^{\otimes_S i} \longrightarrow A_i, \quad a_1 \otimes \dots \otimes a_i \longmapsto a_1 \dots a_i \quad (i \geq 1), \quad \mu_0 = \text{id}_S. \quad (2.3)$$

Definition 2.4. The algebra A is *generated in degree 1 over S* if every μ_i in (2.3) is surjective.

Equivalently, S and A_1 generate A as an algebra.

Example 2.5. The algebra $\mathbb{k}A_n$ of Example 1.9 is generated in degree 1 over $S = \bigoplus_i \mathbb{k}e_{ii}$:

$$e_{ij} = e_{i,i+1}e_{i+1,i+2} \dots e_{j-1,j} \quad (1 \leq i < j \leq n).$$

For the grading $\deg e_{ij} = 2(j-i)$, the same algebra has $A_1 = 0$ and $A_2 \neq 0$, so it is not generated in degree 1.

Tensor algebras and the universal property

Definition 2.6. Let S be a $\mathbb{k}$ -algebra and V an S -bimodule. The *tensor algebra of V over S* is

$$T_S(V) := \bigoplus_{i \in \mathbb{Z}} T_S(V)_i, \quad \text{where} \quad T_S(V)_i = \begin{cases} V^{\otimes_S i}, & i \geq 1, \\ S, & i = 0, \\ 0, & i < 0, \end{cases}$$

with multiplication given by concatenation of tensors and the given S -actions.

The left and right S -actions determine which factors can be concatenated. For $S = \bigoplus_{i=1}^r \mathbb{k}e_i$ with coordinate idempotents,

$$V = \bigoplus_{i,j} e_i V e_j, \quad (e_i V e_j) \otimes_S (e_\ell V e_m) = 0 \quad \text{if } j \neq \ell. \quad (2.4)$$

Indeed, $v \otimes w = v e_j \otimes w = v \otimes e_j w = 0$ when $v \in e_i V e_j$, $w \in e_\ell V e_m$, and $j \neq \ell$.

Example 2.7. Let $S = \mathbb{k}^{\oplus 3}$ with coordinate idempotents e_1, e_2, e_3 , and let $V = \mathbb{k}a \oplus \mathbb{k}b$ have actions $e_1 a e_2 = a$ and $e_2 b e_3 = b$, with all other idempotent actions zero. By (2.4),

$$V \otimes_S V = \mathbb{k}(a \otimes b), \quad V^{\otimes_S 3} = 0, \quad T_S(V) = S \oplus \mathbb{k}a \oplus \mathbb{k}b \oplus \mathbb{k}ab.$$

Here $ab = a \otimes b$. The assignments $e_i \mapsto e_{ii}$, $a \mapsto e_{12}$, and $b \mapsto e_{23}$ give $T_S(V) \cong \mathbb{k}A_3$; the tensor ab maps to e_{13} .

Proposition 2.8 (Universal property of the tensor algebra). Let $\iota_S : S \rightarrow T_S(V)$ and $\iota_V : V \rightarrow T_S(V)$ be the canonical inclusions. Given a unital $\mathbb{k}$ -algebra homomorphism $\varphi : S \rightarrow B$ and an S -bimodule map $f : V \rightarrow B$, where $sbt = \varphi(s)b\varphi(t)$ on B , there is a unique $\mathbb{k}$ -algebra homomorphism $\tilde{f} : T_S(V) \rightarrow B$ making the diagram commute:

$$\begin{array}{ccccc} V & \xrightarrow{\iota_V} & T_S(V) & \xleftarrow{\iota_S} & S \\ & \searrow f & \downarrow \exists! \tilde{f} & \swarrow \varphi & \\ & & B & & \end{array} \quad (2.5)$$

Proof Define $\tilde{f}$ linearly by

$$\tilde{f}(s) = \varphi(s), \quad \tilde{f}(v_1 \otimes_S \cdots \otimes_S v_i) = f(v_1) \cdots f(v_i) \quad (s \in S, i \geq 1). \quad (2.6)$$

The identity $f(svt) = \varphi(s)f(v)\varphi(t)$ makes $\tilde{f}$ well-defined on tensor products and compatible with multiplication. Every homomorphism satisfying (2.5) has the values in (2.6), proving uniqueness. $\square$

Corollary 2.9. *If A is generated in degree 1 over $S = A_0$, there is a surjective graded algebra homomorphism $\mu : T_S(A_1) \twoheadrightarrow A$ whose degree i component is μ_i from (2.3).*

Proof Apply Proposition 2.8 to the inclusions $S \rightarrow A$ and $A_1 \rightarrow A$. Surjectivity of each μ_i gives surjectivity of μ . $\square$

Definition 2.10. *For the map μ in Corollary 2.9, the homogeneous two-sided ideal $I := \text{Ker } \mu$ is the ideal of relations.*

Writing $V = A_1$, we obtain $A \cong T_S(V)/I$ with $I_0 = I_1 = 0$. The bimodule V specifies the generators and their left and right S -actions; I specifies the relations between their products.

Quadratic algebras

Definition 2.11. *Let S be finite-dimensional semisimple, V a finite-dimensional S -bimodule, and $R \subset V \otimes_S V$ an S -sub-bimodule. The algebra $T_S(V)/(R)$ is a quadratic algebra, where (R) denotes the two-sided ideal generated by R .*

Proposition 2.12. *A quadratic algebra $A = T_S(V)/(R)$ has $A_0 = S$ and $A_1 = V$, is locally finite, and is generated in degree 1 over S .*

Proof The ideal (R) has no components in degrees 0 and 1. Each A_i is a quotient of the finite-dimensional space $V^{\otimes_S i}$, and multiplication maps $V^{\otimes_S i}$ onto A_i . $\square$

Example 2.13. *Let $S = \bigoplus_{i=1}^n \mathbb{k}e_{ii}$ and $V = \bigoplus_{i=1}^{n-1} \mathbb{k}e_{i,i+1}$ inside $\mathbb{k}A_n$. By (2.4), the basis tensors of positive length are*

$$e_{i,i+1} \otimes_S e_{i+1,i+2} \otimes_S \cdots \otimes_S e_{j-1,j} \mapsto e_{ij} \quad (1 \leq i < j \leq n).$$

Thus multiplication gives $T_S(V) \cong \mathbb{k}A_n$, a quadratic algebra with $R = 0$.

Example 2.14. *For S, V, a, b from Example 2.7, the quotient*

$$T_S(V)/(ab) \cong \mathbb{k}A_3/(e_{13})$$

is quadratic with $R = \mathbb{k}(a \otimes b)$. Its basis is e_1, e_2, e_3, a, b , and $h_A(t) = 3 + 2t$, as calculated in Example 1.23.

Example 2.15. *Let $S = \mathbb{k}^{\oplus 2}$ with coordinate idempotents e_1, e_2 , and let $V = \mathbb{k}a \oplus \mathbb{k}b$ satisfy $e_1 a e_2 = a$ and $e_2 b e_1 = b$, with all other idempotent actions zero. In $T_S(V)$ only alternating words in a, b survive. The quadratic algebra $A = T_S(V)/(ab)$ has*

$$A = S \oplus \mathbb{k}a \oplus \mathbb{k}b \oplus \mathbb{k}ba, \quad h_A(t) = 2 + 2t + t^2.$$

Every alternating word of length at least 3 contains ab , so all higher components vanish.

Example 2.16. *Let $B = \mathbb{k}A_4$ and $A = B/(e_{14})$, with the grading of Example 1.9. The ideal $I = (e_{14}) = \mathbb{k}e_{14}$ has $I_2 = 0$ but $I_3 \neq 0$. Since $e_{14} = e_{12}e_{23}e_{34}$, the algebra A is generated in degree 1 with a relation of degree 3 and no relations of degree 2. Hence A is not quadratic.*

Example 2.17 (Polynomial and exterior algebras). For the classical comparison, take $S = \mathbb{k}$ and $V = \bigoplus_{i=1}^n \mathbb{k}x_i$. The *symmetric algebra* and *exterior algebra* are

$$\begin{aligned} P &:= \text{Sym}(V) = T_{\mathbb{k}}(V)/(x_i x_j - x_j x_i \mid i < j) \cong \mathbb{k}[x_1, \dots, x_n], \\ E &:= \bigwedge V = T_{\mathbb{k}}(V)/(x_i^2, x_i x_j + x_j x_i \mid i < j). \end{aligned} \quad (2.7)$$

Both are quadratic. The polynomial basis allows repetitions; the exterior basis consists of 1 and $x_{i_1} \cdots x_{i_d}$ with $i_1 < \cdots < i_d$. Consequently

$$h_P(t) = \frac{1}{(1-t)^n}, \quad h_E(t) = (1+t)^n.$$

The relations for E are equivalent to $v^2 = 0$ for every $v \in V$, in every characteristic.

The degree 2 relations of a quotient

Let $A = T_S(V)/I$ be generated in degree 1, with $I_0 = I_1 = 0$ as in Definition 2.10. In degree 2,

$$A_2 \cong (V \otimes_S V)/I_2, \quad \text{where } I_2 = I \cap (V \otimes_S V).$$

Definition 2.18. The *quadratic part* of A is $qA := T_S(V)/(I_2)$.

The inclusion $(I_2) \subset I$ induces a surjective graded homomorphism $\pi_A : qA \twoheadrightarrow A$, and

$$\pi_A \text{ is an isomorphism} \iff I = (I_2) \iff A \text{ is quadratic.}$$

Example 2.19. For $A = \mathbb{k}A_4/(e_{14})$ from Example 2.16, we have $I_2 = 0$, so

$$qA = \mathbb{k}A_4, \quad \pi_A : \mathbb{k}A_4 \twoheadrightarrow \mathbb{k}A_4/(e_{14}), \quad \text{Ker } \pi_A = \mathbb{k}e_{14}.$$

Thus qA does not impose the cubic relation $e_{12}e_{23}e_{34} = 0$.

Orthogonal complements and quadratic duality

Definition 2.20. For an S -sub-bimodule U of a finite-dimensional S -bimodule M , define the *right orthogonal complement* and *left orthogonal complement* by

$$U^\perp := \{f \in M^* \mid f(U) = 0\}, \quad {}^\perp U := \{g \in {}^*M \mid g(U) = 0\}.$$

Both $U^\perp$ and ${}^\perp U$ are S -sub-bimodules by (1.2). Fix a symmetrising form τ as in Definition 1.37. For the comparison maps $\alpha_M(f) = \tau \circ f$ and $\beta_M(g) = \tau \circ g$ in (1.3),

$$\alpha_M(U^\perp) = \beta_M({}^\perp U) = \{\lambda \in D(M) \mid \lambda(U) = 0\}. \quad (2.8)$$

Indeed, if $\alpha_M(f)(U) = 0$, then

$$\tau(sf(u)) = \tau(f(su)) = 0 \quad \text{for all } s \in S, u \in U,$$

so non-degeneracy of τ gives $f(U) = 0$. For β_M , use $\tau(g(u)s) = \tau(g(us))$ instead.

For $A = T_S(V)/(R)$, the map θ_ℓ in (2.1) identifies $R^\perp$ with a sub-bimodule of $V^* \otimes_S V^*$. These are the generators and relations used for the quadratic dual in Lecture 3:

$$V^* = \text{Hom}_S(V, S), \quad R^\perp \subset V^* \otimes_S V^*.$$

The generators V^* are assigned degree 1. In contrast, the graded S -dual $V^{(*)}$ from Definition 1.40 has degree -1 when $V = A_1$.

Example 2.21. For the pair $P = \text{Sym}(V)$ and $E = \bigwedge V$ in Example 2.17, take $V = \mathbb{k}x \oplus \mathbb{k}y$. The relation spaces are

$$\begin{aligned} R_P &= \mathbb{k}(x \otimes y - y \otimes x), \\ R_E &= \mathbb{k}(x \otimes x) \oplus \mathbb{k}(y \otimes y) \oplus \mathbb{k}(x \otimes y + y \otimes x). \end{aligned}$$

Let $x^*, y^* \in V^*$ be the coordinate functionals. With the tensor evaluation θ_ℓ in (2.2),

$$\begin{aligned} R_P^\perp &= \text{span}_{\mathbb{k}}\{x^* \otimes x^*, y^* \otimes y^*, x^* \otimes y^* + y^* \otimes x^*\}, \\ R_E^\perp &= \mathbb{k}(x^* \otimes y^* - y^* \otimes x^*). \end{aligned}$$

Thus the quadratic dual presentations exchange the symmetric and exterior relations.

Exercise 2.22. Let $A = \mathbb{k}A_4/(e_{13}, e_{24})$ with the grading of Example 1.9. Find a homogeneous basis, compute $h_A(t)$, and write $A = T_S(V)/(R)$ with explicit S, V, R .

Exercise 2.23. For the bimodule V in Example 2.15, find a basis of each $T_S(V)_d$ and show that $h_{T_S(V)}(t) = 2/(1-t)$. Compare with the quotient $T_S(V)/(ab)$.

Exercise 2.24. For $P = \text{Sym}(V)$ and $E = \bigwedge V$ in Example 2.17, prove

$$\dim_{\mathbb{k}} P_2 = \frac{n(n+1)}{2}, \quad \dim_{\mathbb{k}} E_2 = \frac{n(n-1)}{2} \quad \text{where } n = \dim_{\mathbb{k}} V.$$

References

- [BGS96] A. Beilinson, V. Ginzburg, and W. Soergel, *Koszul duality patterns in representation theory*, J. Amer. Math. Soc. 9 (1996), 473–527.
- [GI20] J. Grant and O. Iyama, *Higher preprojective algebras, Koszul algebras, and superpotentials*, Compos. Math. 156 (2020), 2588–2627.